CYBERSECURITY VULNERABILITY ASSESSMENT & REMEDIATION REPORT

Target System: Linux Server

Prepared For: Technical Team & SOC Manager

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Date: March 2026

1. Executive Summary

A comprehensive vulnerability assessment was conducted on a Linux-based web server hosting an Nginx application. The engagement followed a structured vulnerability management lifecycle including credentialed scanning, web application scan, vulnerability analysis, automated remediation, and post-remediation validation.

The credentialed scan using SSH authentication did not return any vulnerability indicating potential scan scope limitations. However, the unauthenticated web application scan identified multiple vulnerabilities affecting the Nginx service, including high-risk issues such as remote code execution and denial-of-service conditions.

Automated remediation using Ansible successfully upgraded Nginx from version **1.15.5 to 1.24.0**, eliminating all identified vulnerabilities. Post-remediation validation confirmed no remaining detectable vulnerabilities within the scan scope.

While the system shows a significantly improved security posture, it is important to note that no system can be considered completely secure. Residual risks may exist outside the scope of this assessment.

2. Scope of Engagement

The assessment covered the following:

- Credentialed vulnerability scan of Linux server via SSH
- Web application vulnerability scan (unauthenticated)
- Identification and analysis of Nginx-related vulnerabilities
- Risk scoring using CVSS, VPR, and EPSS metrics
- Exploitability and impact assessment
- Automated remediation using Ansible
- SMTP configuration for automated reporting
- Post-patch validation scanning

3. Environment Overview

Component	Details
Operating System	Ubuntu Linux
Web Server	Nginx
Initial Version	1.15.5
Updated Version	1.24.0
Services	SSH
Scanner	Nessus Essentials
Automation Tool	Ansible
Reporting	Gmail SMTP

4. Methodology

4.1 Preparation

- Installed and configured Nessus Essentials
- Enabled and configured SSH for credentialed scanning
- Deployed vulnerable Nginx version (1.15.5)
- Installed Ansible and created patch automation playbook
- Configured SMTP for automated reporting

4.2 Credentialed Scan Configuration

- Authentication Method: SSH (Username/Password)
- Username: root
- Privilege Escalation: su
- SSH Version: Verified using ssh -V

Credentialed scanning was configured to allow deeper inspection of system-level vulnerabilities.

4.3 Scanning Activities

- Executed credentialed scan on Linux host
- Executed web application scan using default Nessus configuration
- Collected vulnerability data and supporting evidence

4.4 Analysis

- Mapped vulnerabilities to CVE identifiers
- Analysed CVSS, VPR, and EPSS scores
- Assessed exploit availability and ease
- Evaluated operational and business impact

4.5 Remediation

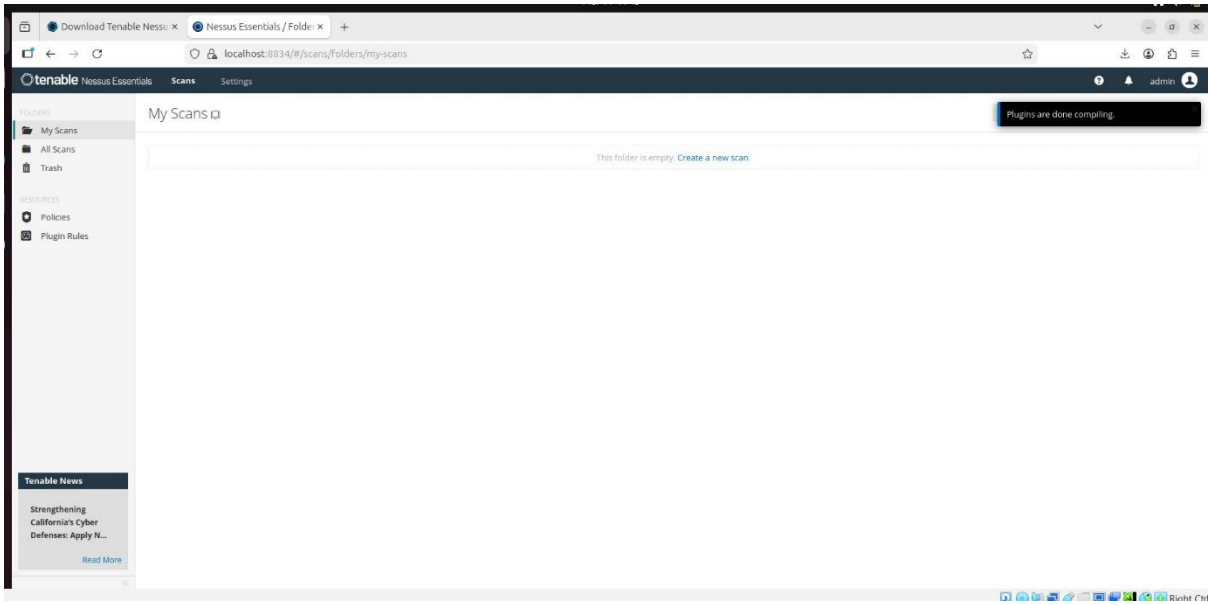
- Developed and executed Ansible playbook
- Upgraded Nginx to latest stable version
- Restarted and validated service

4.6 Validation

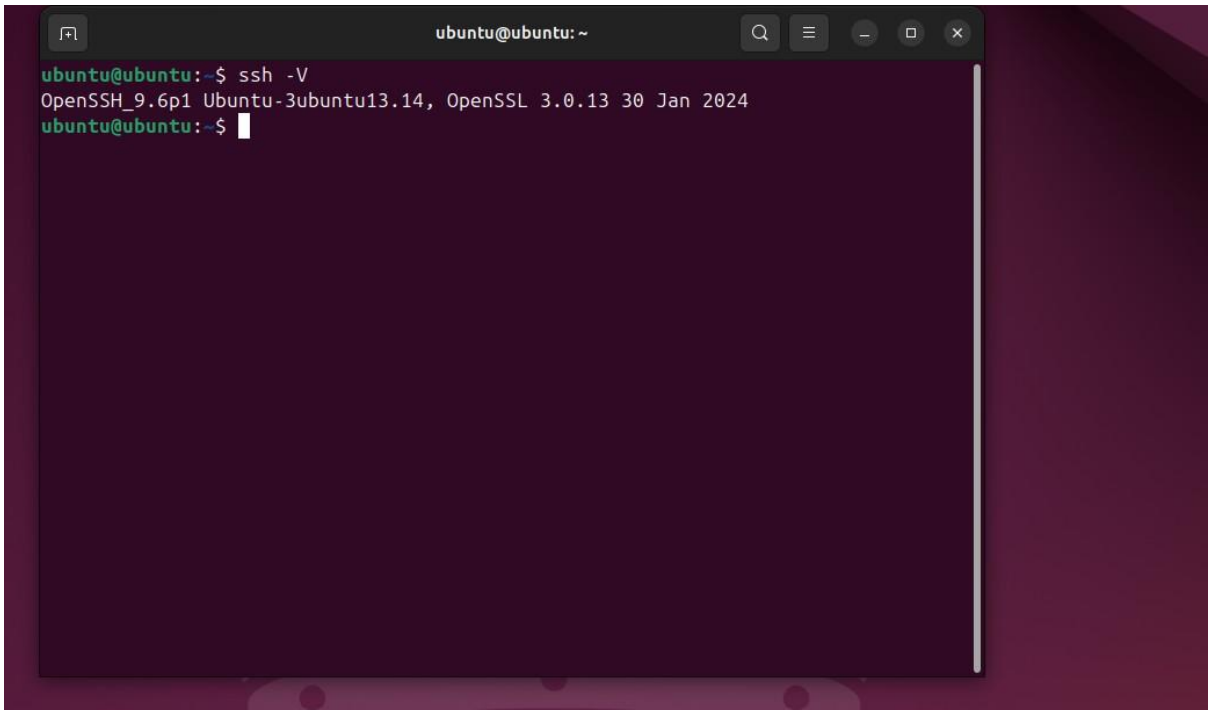
- Re-ran vulnerability scans
- Verified remediation effectiveness

Installation/Configuration Screenshots

Nessus Dashboard



SSH Installation



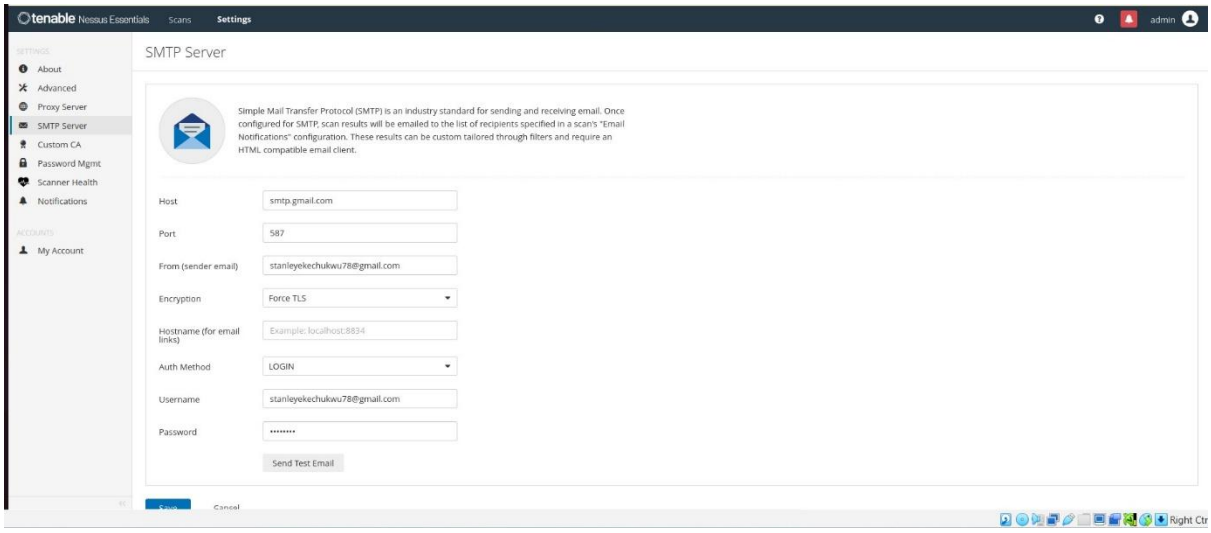
Nginx after compilation

```
ubuntu@ubuntu: ~/nginx-1.15.5
                '/usr/local/nginx/sbin/nginx.old'
cp objs/nginx '/usr/local/nginx/sbin/nginx'
test -d '/usr/local/nginx/conf' \
        || mkdir -p '/usr/local/nginx/conf'
cp conf/koi-win '/usr/local/nginx/conf'
cp conf/koi-utf '/usr/local/nginx/conf'
cp conf/win-utf '/usr/local/nginx/conf'
test -f '/usr/local/nginx/conf/mime.types' \
        || cp conf/mime.types '/usr/local/nginx/conf'
cp conf/mime.types '/usr/local/nginx/conf/mime.types.default'
test -f '/usr/local/nginx/conf/fastcgi_params' \
        || cp conf/fastcgi_params '/usr/local/nginx/conf'
cp conf/fastcgi_params \
        '/usr/local/nginx/conf/fastcgi_params.default'
test -f '/usr/local/nginx/conf/fastcgi.conf' \
        || cp conf/fastcgi.conf '/usr/local/nginx/conf'
cp conf/fastcgi.conf '/usr/local/nginx/conf/fastcgi.conf.default'
test -f '/usr/local/nginx/conf/uwsgi_params' \
        || cp conf/uwsgi_params '/usr/local/nginx/conf'
cp conf/uwsgi_params \
        '/usr/local/nginx/conf/uwsgi_params.default'
test -f '/usr/local/nginx/conf/scgi_params' \
        || cp conf/scgi_params '/usr/local/nginx/conf'
cp conf/scgi_params \
        '/usr/local/nginx/conf/scgi_params.default'
test -f '/usr/local/nginx/conf/nginx.conf' \
        || cp conf/nginx.conf '/usr/local/nginx/conf/nginx.conf'
cp conf/nginx.conf '/usr/local/nginx/conf/nginx.conf.default'
test -d '/usr/local/nginx/logs' \
        || mkdir -p '/usr/local/nginx/logs'
test -d '/usr/local/nginx/logs' \
        || mkdir -p '/usr/local/nginx/logs'
test -d '/usr/local/nginx/html' \
        || cp -R html '/usr/local/nginx'
test -d '/usr/local/nginx/logs' \
        || mkdir -p '/usr/local/nginx/logs'
make[1]: Leaving directory '/home/ubuntu/nginx-1.15.5'
ubuntu@ubuntu:~/nginx-1.15.5$
```

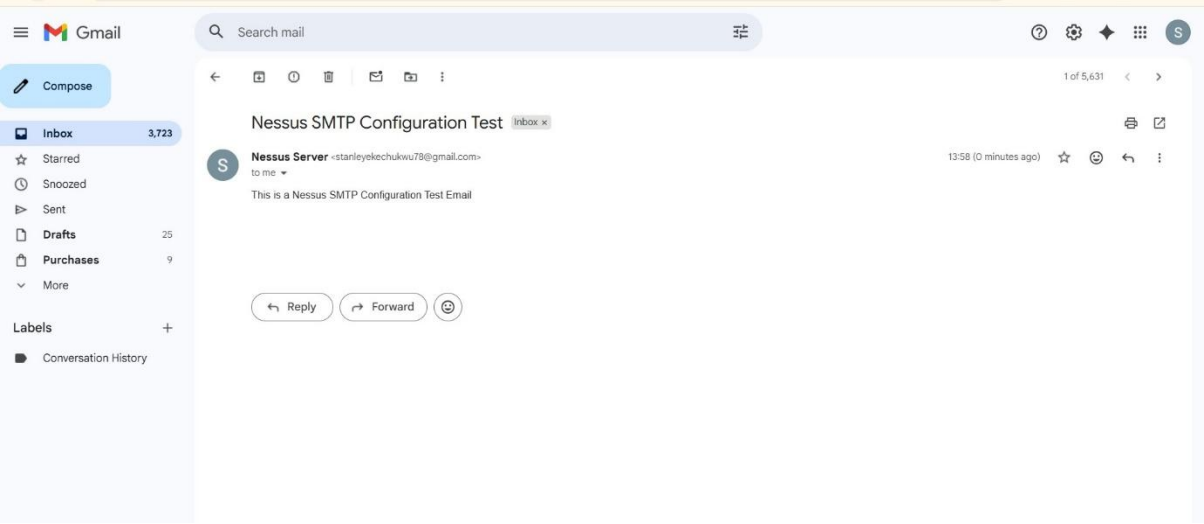
Nginx Successfully Installed, indicating Nginx version (1.15.5)

```
ubuntu@ubuntu:~/nginx-1.15.5$ sudo /usr/local/nginx/sbin/nginx
ubuntu@ubuntu:~/nginx-1.15.5$ curl http://localhost
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
    body {
        width: 35em;
        margin: 0 auto;
        font-family: Tahoma, Verdana, Arial, sans-serif;
    }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
ubuntu@ubuntu:~/nginx-1.15.5$ sudo /usr/local/nginx/sbin/nginx -t
nginx: the configuration file /usr/local/nginx/conf/nginx.conf syntax is ok
nginx: configuration file /usr/local/nginx/conf/nginx.conf test is successful
ubuntu@ubuntu:~/nginx-1.15.5$ /usr/local/nginx/sbin/nginx -v
nginx version: nginx/1.15.5
ubuntu@ubuntu:~/nginx-1.15.5$
```

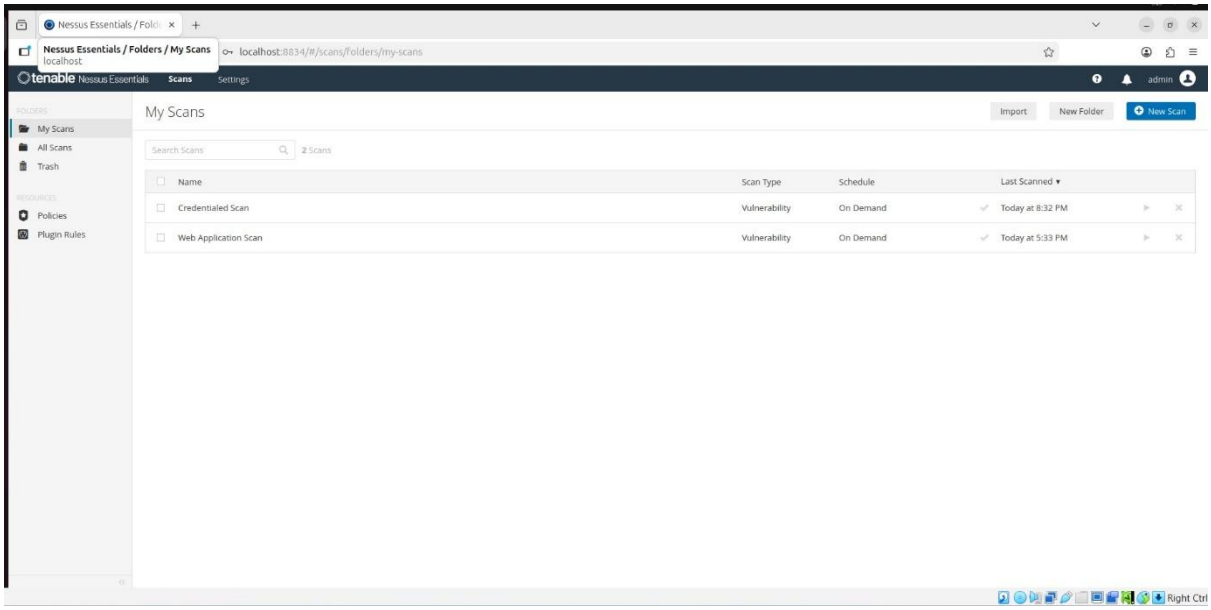
SMTP Configuration in Nessus



SMTP Automated Email Confirmation



Completed Scans: Credentialed Scan and Web Application Scan



5. Credentialed Scan Results

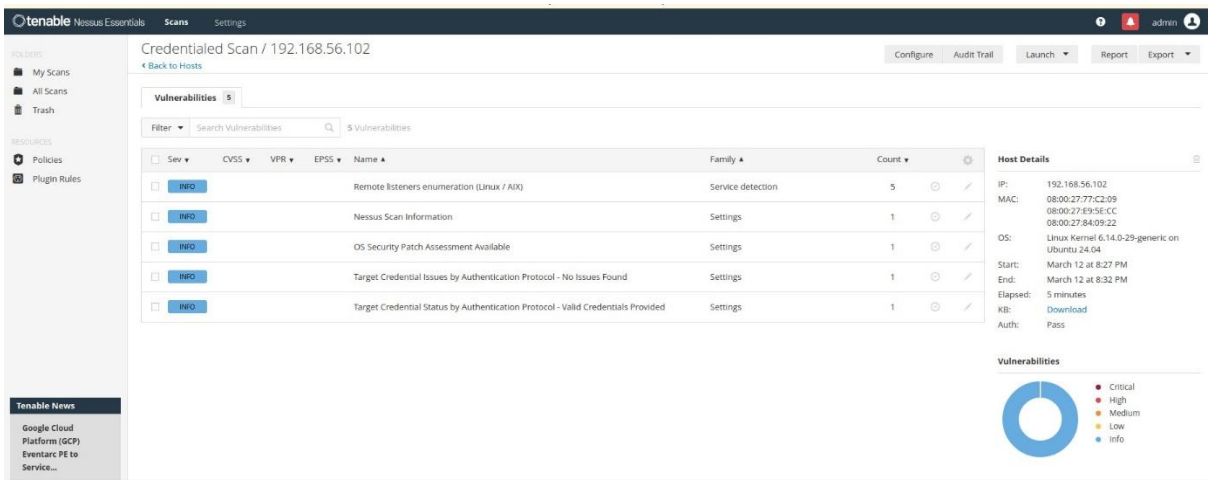
The credentialed scan returned no vulnerabilities.

Interpretation:

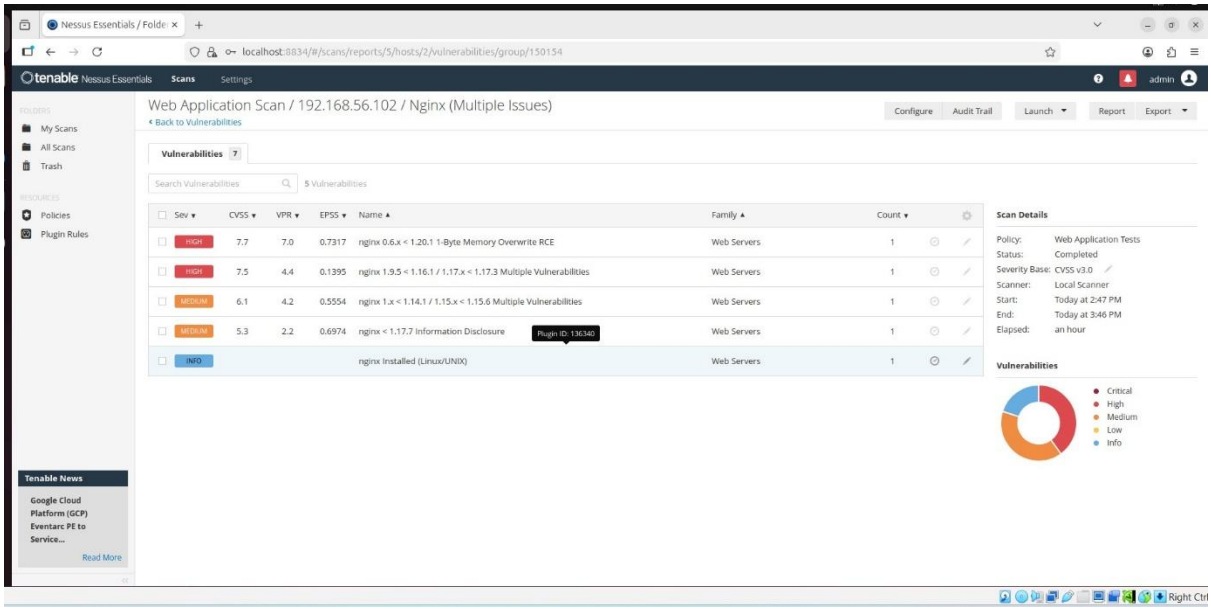
- The system may be well-hardened at the OS level
- Alternatively, scan configuration or scope limitations may have restricted findings

This highlights the importance of combining credentialed and application-layer scans for full visibility.

Details of Credentialed Scan showing no vulnerabilities



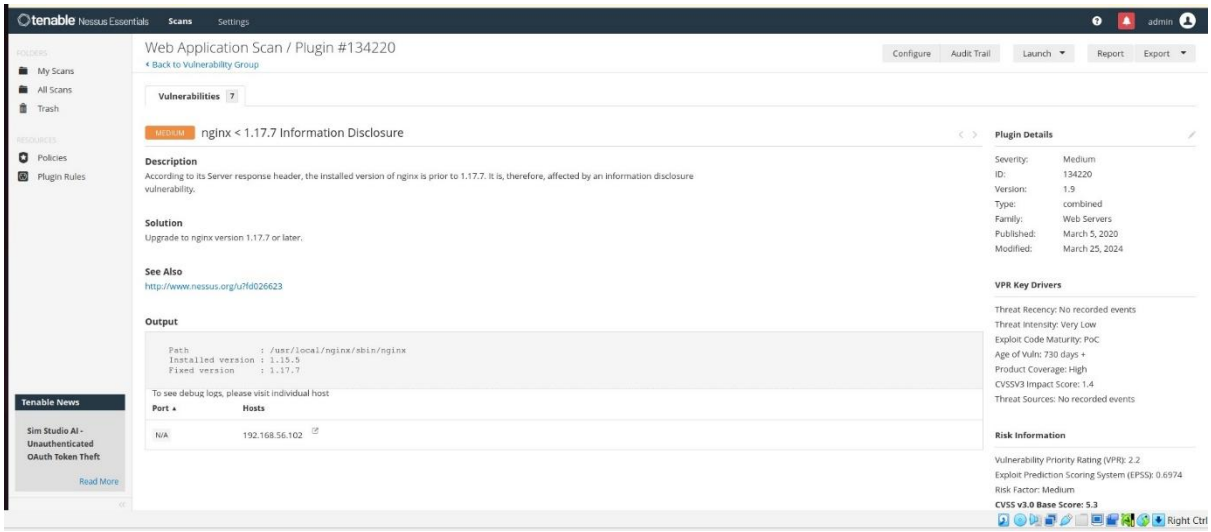
Details of Web Application Scan Showing 4 Parent Vulnerabilities



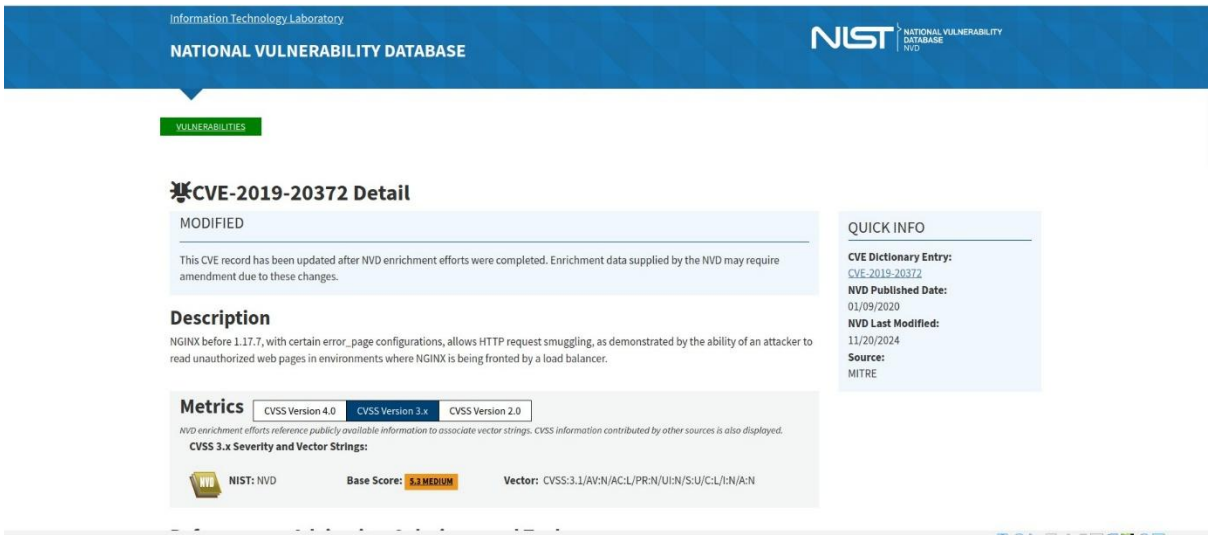
6. Vulnerability Findings (Pre-Remediation)

Below are the four main vulnerabilities, including their sub-vulnerabilities, grouped exactly as Nessus reported them. Findings have been analysed using Nessus intelligence and NIST (National Institute of Standards and Technology) NVD (National Vulnerability Database) intelligence:

A. Nginx < 1.17.7 Information Disclosure



Further Details on NIST NVD Using CVE Tag



Nessus Description: According to its Server response header, the installed version of nginx is prior to 1.17.7. It is therefore, affected by an information disclosure vulnerability.

NIST Description: NGINX before 1.17.7, with certain error_page configurations, allows HTTP request smuggling, as demonstrated by the ability of an attacker to read unauthorised web pages in environments where Nginx is being fronted by a load balancer.

Vulnerability Publication Date: 24th December, 2019.
Patch Publication Date: 24th December, 2019.

Exploitation Method: This vulnerability can be exploited through HTTP request smuggling, where an attacker crafts a request that is interpreted differently by the load balancer and Nginx. As a result, the attacker can bypass the load balancer's security controls and send unauthorised requests directly to the backend server. This may allow the attacker to access restricted web pages, interfere with other users' requests, or gather information about internal server behaviour. While the vulnerability does not directly impact integrity or availability, it poses a confidentiality risk and can be used as part of a broader attack chain.

Exploit Available - True
Exploit Ease – True
Vulnerability ID as assigned by Nessus: 134220
Severity according to Nessus: Medium
CVE(Common Vulnerabilities and Exposures) Code: CVE-2019-20372

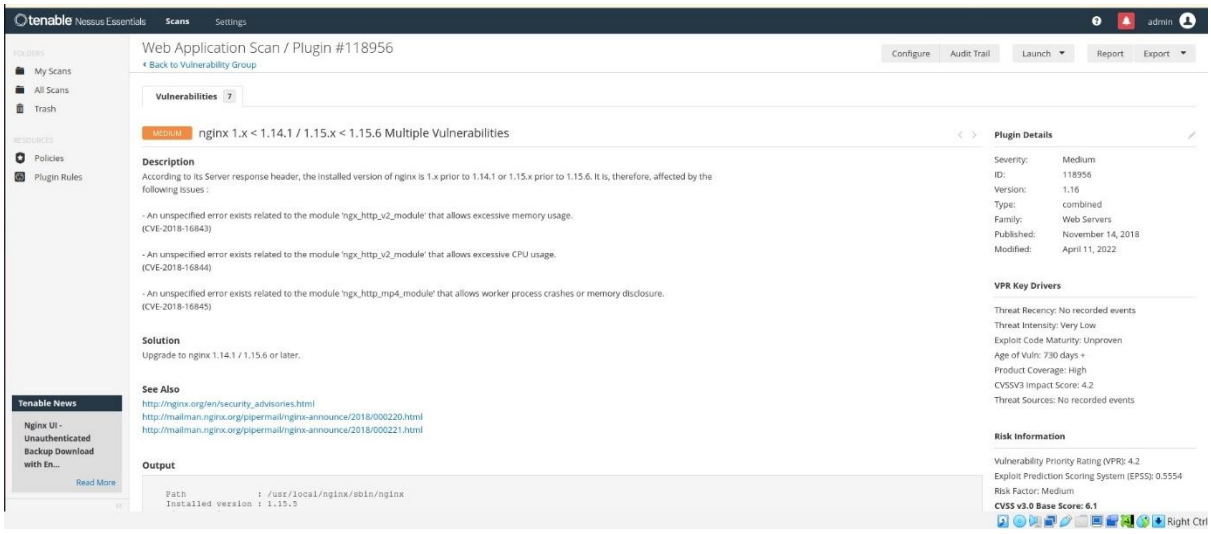
This vulnerability has a **CVSSV3(Common Vulnerability Scoring System) Impact Score of 1.4** which indicates that the vulnerability has minimal effect on confidentiality, integrity, and availability. Even if exploited, the resulting impact on the system would be low, making this primarily an information disclosure issue rather than a direct threat to system security.

This vulnerability has a **CVSS v3.0 Base Score of 5.3** which classifies this vulnerability as Medium severity. It is easily exploitable over the network without authentication and results in low-level information disclosure, but does not impact system integrity or availability. While not immediately dangerous on its own, it increases the attack surface and should be remediated as part of routine patching to prevent its use in reconnaissance or attack chaining.

VPR(Vulnerability Priority Rating): 2.2, this indicates low real-world threat priority based on Tenable's threat intelligence.

EPSS(Exploit Prediction Scoring System) Value: 0.6974, this shows a high likelihood of exploitation, meaning attackers frequently target this type of weakness.

B. Nginx 1.x < 1.14.1 / 1.15.x < 1.15.6 Multiple Vulnerabilities



Sub vulnerability1

NATIONAL VULNERABILITY DATABASE

NIST NATIONAL VULNERABILITY DATABASE NVD

VULNERABILITIES

CVE-2018-16843 Detail

MODIFIED

This CVE record has been updated after NVD enrichment efforts were completed. Enrichment data supplied by the NVD may require amendment due to these changes.

Description

nginx before versions 1.15.6 and 1.14.1 has a vulnerability in the implementation of HTTP/2 that can allow for excessive memory consumption. This issue affects nginx compiled with the ngx_http_v2_module (not compiled by default) if the 'http2' option of the 'listen' directive is used in a configuration file.

QUICK INFO

CVE Dictionary Entry:
CVE-2018-16843
NVD Published Date:
11/07/2018
NVD Last Modified:
11/20/2024
Source:
Red Hat, Inc.

Metrics CVSS Version 4.0 | CVSS Version 3.x | CVSS Version 2.0

NVD enrichment efforts reference publicly available information to associate vector strings. CVSS information contributed by other sources is also displayed.

CVSS 3.x Severity and Vector Strings:

NIST: NVD **Base Score:** 7.5 HIGH **Vector:** CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:N/I:N/A:H

CNA: Red Hat, Inc. **Base Score:** 5.3 MEDIUM **Vector:** CVSS:3.0/AV:N/AC:L/PR:N/UI:N/S:U/C:N/I:N/A:L

Right Ctrl

Sub vulnerability2

NATIONAL VULNERABILITY DATABASE

NIST NATIONAL VULNERABILITY DATABASE NVD

VULNERABILITIES

CVE-2018-16844 Detail

MODIFIED

This CVE record has been updated after NVD enrichment efforts were completed. Enrichment data supplied by the NVD may require amendment due to these changes.

Description

nginx before versions 1.15.6 and 1.14.1 has a vulnerability in the implementation of HTTP/2 that can allow for excessive CPU usage. This issue affects nginx compiled with the ngx_http_v2_module (not compiled by default) if the 'http2' option of the 'listen' directive is used in a configuration file.

QUICK INFO

CVE Dictionary Entry:
CVE-2018-16844
NVD Published Date:
11/07/2018
NVD Last Modified:
11/20/2024
Source:
Red Hat, Inc.

Metrics CVSS Version 4.0 | CVSS Version 3.x | CVSS Version 2.0

NVD enrichment efforts reference publicly available information to associate vector strings. CVSS information contributed by other sources is also displayed.

CVSS 3.x Severity and Vector Strings:

NIST: NVD **Base Score:** 7.5 HIGH **Vector:** CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:N/I:N/A:H

CNA: Red Hat, Inc. **Base Score:** 5.3 MEDIUM **Vector:** CVSS:3.0/AV:N/AC:L/PR:N/UI:N/S:U/C:N/I:N/A:L

Right Ctr

Sub vulnerability3

NATIONAL VULNERABILITY DATABASE

NIST NATIONAL VULNERABILITY DATABASE NVD

VULNERABILITIES

CVE-2018-16845 Detail

MODIFIED

This CVE record has been updated after NVD enrichment efforts were completed. Enrichment data supplied by the NVD may require amendment due to these changes.

Description

nginx before versions 1.15.6, 1.14.1 has a vulnerability in the ngx_http_mp4_module, which might allow an attacker to cause infinite loop in a worker process, cause a worker process crash, or might result in worker process memory disclosure by using a specially crafted mp4 file. The issue only affects nginx if it is built with the ngx_http_mp4_module (the module is not built by default) and the .mp4. directive is used in the configuration file. Further, the attack is only possible if an attacker is able to trigger processing of a specially crafted mp4 file with the ngx_http_mp4_module.

QUICK INFO

CVE Dictionary Entry:
CVE-2018-16845
NVD Published Date:
11/07/2018
NVD Last Modified:
11/20/2024
Source:
Red Hat, Inc.

Metrics CVSS Version 4.0 | CVSS Version 3.x | CVSS Version 2.0

NVD enrichment efforts reference publicly available information to associate vector strings. CVSS information contributed by other sources is also displayed.

CVSS 3.x Severity and Vector Strings:

NIST: NVD **Base Score:** 6.1 MEDIUM **Vector:** CVSS:3.1/AV:L/AC:L/PR:N/UI:R/S:U/C:L/I:N/A:H

Right Ctrl

Nessus Description: According to its Server response header, the installed version of nginx is 1.x prior to 1.14.1 or 1.15.x prior to 1.15.6. It is, therefore, affected by the following issues:
I. An unspecified error exists related to the module 'ngx_http_v2_module' that allows excessive memory usage. **(CVE-2018-16843)**

II. An unspecified error exists related to the module 'ngx_http_v2_module' that allows excessive CPU usage. **(CVE-2018-16844)**

III. An unspecified error exists related to the module 'ngx_http_mp4_module' that allows worker process crashes or memory disclosure. **(CVE-2018-16845)**

NIST Description:

CVE-2018-16843 - nginx before versions 1.15.6 and 1.14.1 has a vulnerability in the implementation of HTTP/2 that can allow for excessive memory consumption. This issue affects nginx compiled with the ngx_http_v2_module (not compiled by default) if the 'http2' option of the 'listen' directive is used in a configuration file.

CVE-2018-16844 – nginx before versions 1.15.6 and 1.14.1 has a vulnerability in the implementation of HTTP/2 that can allow for excessive CPU usage. This issue affects nginx compiled with the ngx_http_v2_module (not compiled by default) if the 'http2' option of the 'listen' directive is used in a configuration file.

CVE-2018-16845 – nginx before versions 1.15.6, 1.14.1 has a vulnerability in the ngx_http_mp4_module, which might allow an attacker to cause infinite loop in a worker process, cause a worker process crash, or might result in worker process memory disclosure by using a specially crafted mp4 file. The issue only affects nginx if it is built with the ngx_http_mp4_module (the module is not built by default) and the .mp4. directive is used in the configuration file. Further, the attack is only possible if an attacker is able to trigger processing of a specially crafted mp4 file with the ngx_http_mp4_module.

Vulnerability Publication Date: 6th January, 2018.
Patch Publication Date: 6th November, 2018.

Exploitation Method: These three sub vulnerabilities affecting older Nginx versions can be exploited through malformed HTTP/2 or MP4 requests, depending on the modules enabled in the server configuration. **CVE-2018-16843** and **CVE-2018-16844** affect the ngx_http_v2_module when HTTP/2 is enabled, allowing an attacker to send specially crafted HTTP/2 requests that trigger excessive memory allocation or excessive CPU usage. These conditions can significantly degrade performance or lead to denial-of-service. **CVE-2018-16845** affects the ngx_http_mp4_module when the .mp4 directive is used and can be exploited by submitting a specially crafted MP4 file. This may cause an infinite loop, crash a worker process, or expose small fragments of memory. Collectively, these vulnerabilities primarily threaten availability, with limited confidentiality exposure, and can be abused to disrupt service or gather minor internal information about the server.

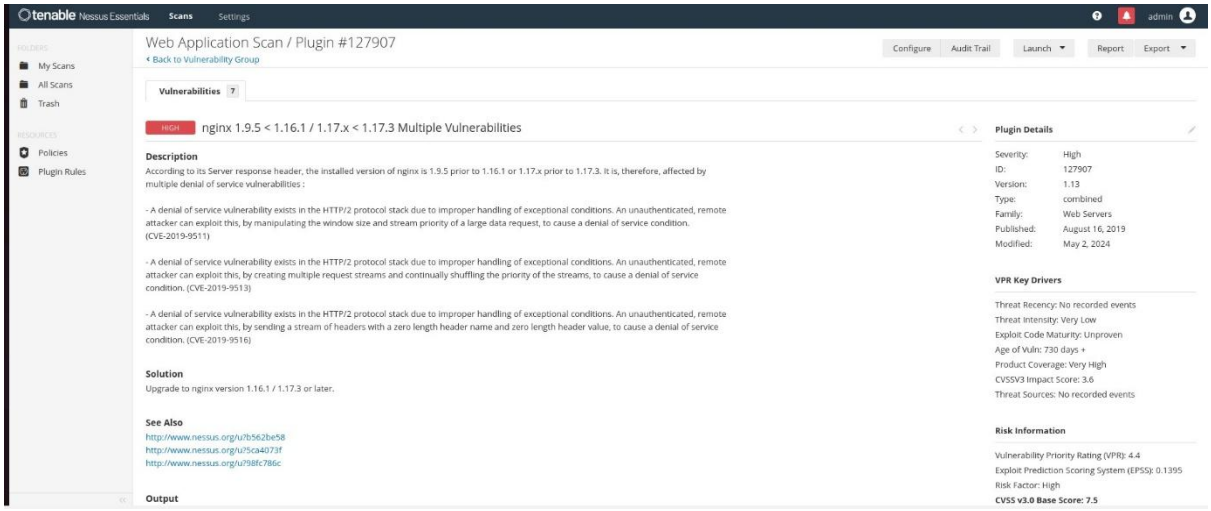
Exploit Ease – No known exploits are available.
Vulnerability ID as assigned by Nessus: 118956
Severity according to Nessus: Medium
CVE Code: CVE-2018-16843, CVE-2018-16844, CVE-2018-16845.

The **CVSS v3.0 Base Scores for these vulnerabilities range from 6.1 to 7.5**, indicating medium to high technical severity, primarily due to their high availability impact. The **CVSSV3 impact metrics of 4.2** confirm that these issues do not affect confidentiality or integrity in a significant way, but can cause substantial service disruption through excessive memory usage, excessive CPU consumption, or worker process crashes.

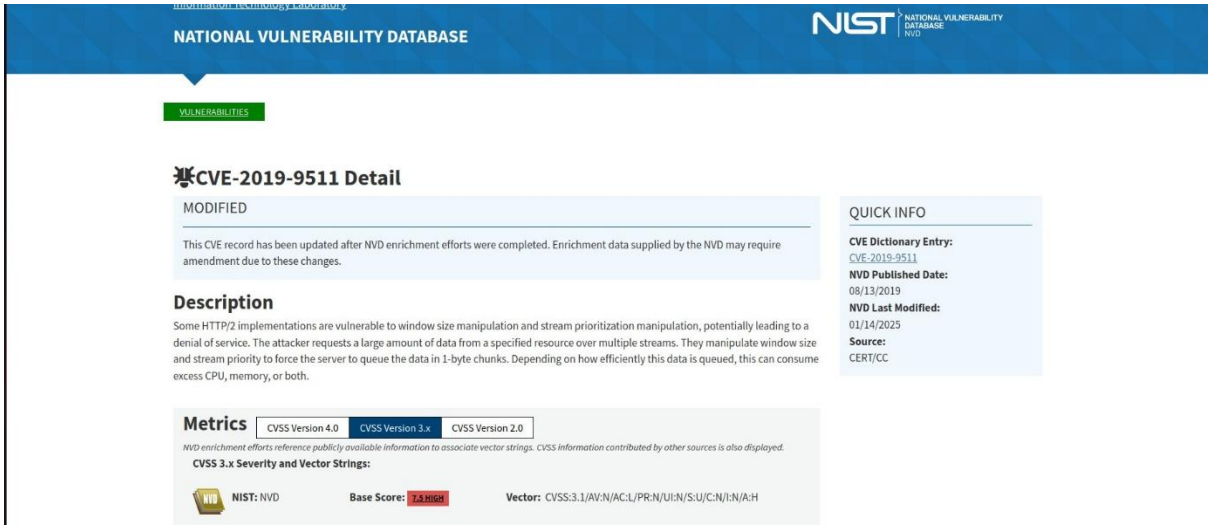
The **VPR score of 4.2** suggests a medium real-world threat level, with no known exploits and low observed attacker activity.

EPSS value of 0.5554 indicates a moderate likelihood of exploitation, meaning these vulnerabilities may still be targeted by automated scanning tools or opportunistic attackers. Overall, the vulnerabilities pose a meaningful availability risk and should be remediated as part of routine patching.

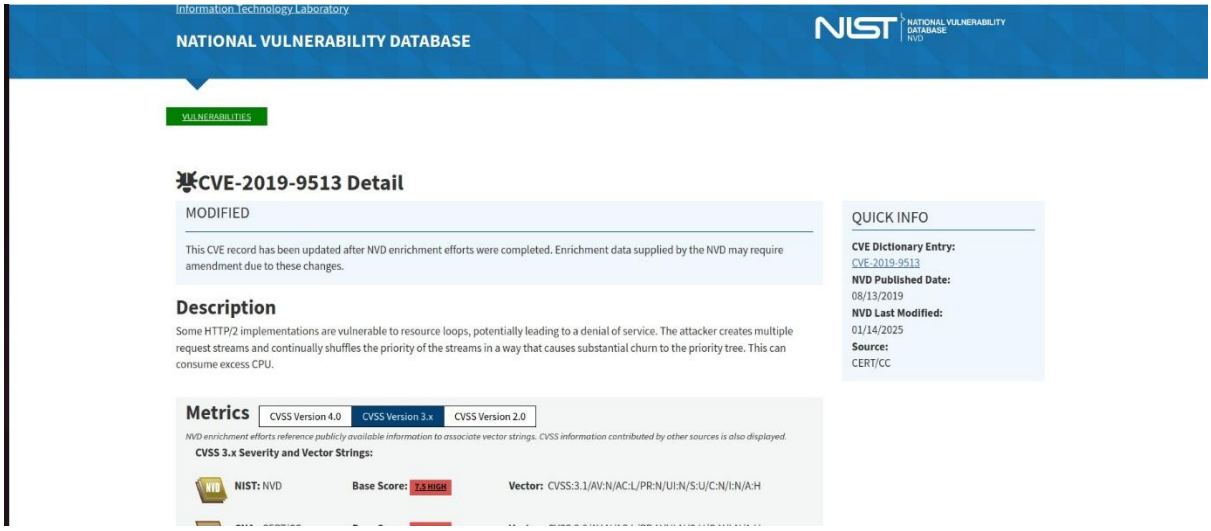
C. Nginx 1.9.5 < 1.16.1 / 1.17.x < 1.17.3 Multiple Vulnerabilities



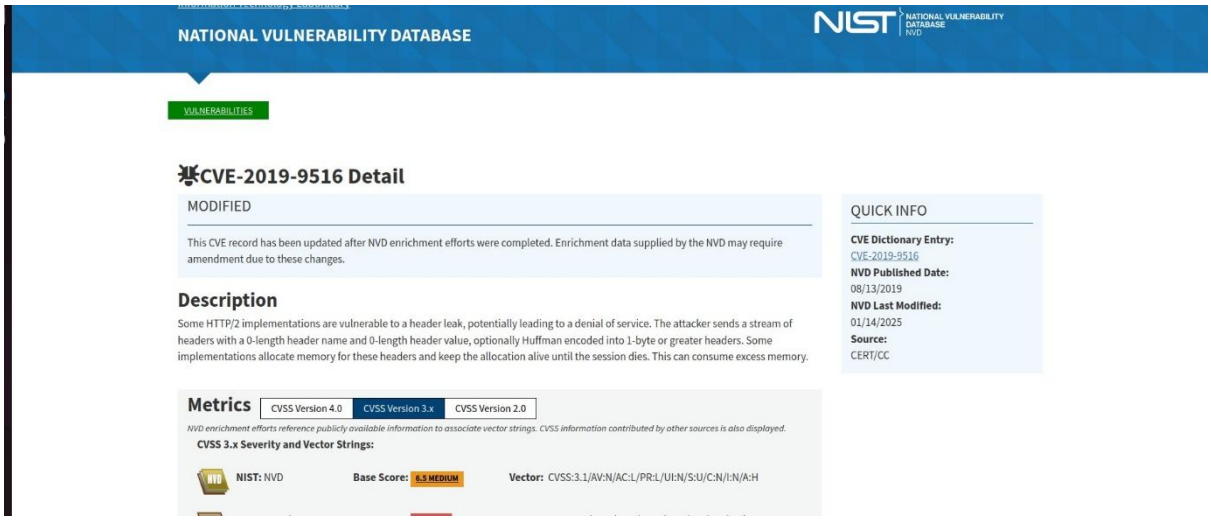
Sub vulnerability1



Sub vulnerability2



Sub vulnerability3



Nessus Description: According to its Server response header, the installed version of nginx is 1.9.5 prior to 1.16.1 or 1.17.x prior to 1.17.3. It is, therefore, affected by multiple denial of service vulnerabilities:
I. A denial of service vulnerability exists in the HTTP/2 protocol stack due to improper handling of exceptional conditions. An unauthenticated, remote attacker can exploit this, by manipulating the window size and stream priority of a large data request, to cause a denial of service condition. **(CVE-2019-9511)**

II. A denial of service vulnerability exists in the HTTP/2 protocol stack due to improper handling of exceptional conditions. An unauthenticated, remote attacker can exploit this, by creating multiple request streams and continually shuffling the priority of the streams, to cause a denial of service condition. **(CVE-2019-9513)**

III. A denial of service vulnerability exists in the HTTP/2 protocol stack due to improper handling of exceptional conditions. An unauthenticated, remote attacker can exploit this, by sending a stream of headers with a zero length header name and zero length header value, to cause a denial of service condition. **(CVE-2019-9516)**

NIST Description:

CVE-2019-9511 - Some HTTP/2 implementations are vulnerable to window size manipulation and stream prioritization manipulation, potentially leading to a denial of service. The attacker requests a large amount of data from a specified resource over multiple streams. They manipulate window size and stream priority to force the server to queue the data in 1-byte chunks. Depending on how efficiently this data is queued, this can consume excess CPU, memory, or both.

CVE-2019-9513 – Some HTTP/2 implementations are vulnerable to resource loops, potentially leading to a denial of service. The attacker creates multiple request streams and continually shuffles the priority of the streams in a way that causes substantial churn to the priority tree. This can consume excess CPU.

CVE-2019-9516 – Some HTTP/2 implementations are vulnerable to a header leak, potentially leading to a denial of service. The attacker sends a stream of headers with a 0-length header name and 0-length header value, optionally Huffman encoded into 1-byte or greater headers. Some implementations allocate memory for these headers and keep the allocation alive until the session dies. This can consume excess memory.

Vulnerability Publication Date: 13th August, 2019.
Patch Publication Date: 13th August, 2019.

Exploitation Method: These sub vulnerabilities consist of three HTTP/2 denial-of-service issues that can be exploited remotely without authentication. **CVE-2019-9511** can be exploited by manipulating HTTP/2 window sizes and stream priorities, forcing the server to queue response data in extremely small chunks, which consumes excessive CPU and memory. **CVE-2019-9513** involves creating multiple HTTP/2 streams and continually shuffling their priorities, causing heavy churn in the priority tree and resulting in high CPU usage. **CVE-2019-9516** abuses 0-length header names and values, causing the server to allocate and retain memory unnecessarily, leading to memory exhaustion. All three vulnerabilities allow an attacker to degrade performance or cause a denial-of-service condition by overwhelming the server's resource handling mechanisms.

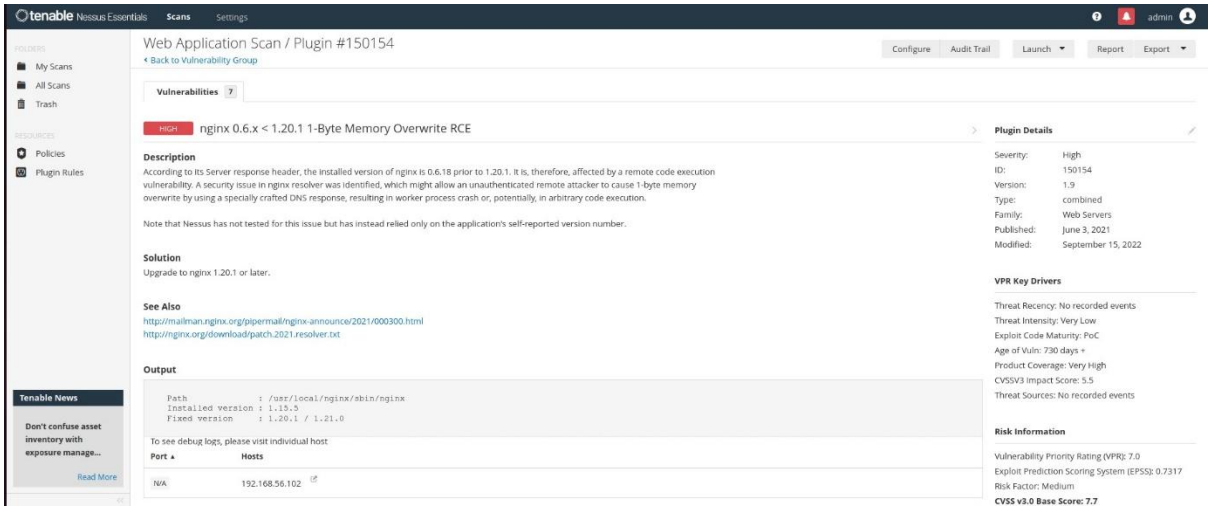
Exploit Available - False
Exploit Ease – No known exploits are available.
Nessus reports that no exploit is available for this vulnerability because there is no publicly released exploit code or weaponised module in common exploit frameworks.

Vulnerability ID as assigned by Nessus: 127907
Severity according to Nessus: High
CVE Code: CVE-2019-9511, CVE-2019-9513, CVE-2019-9516.

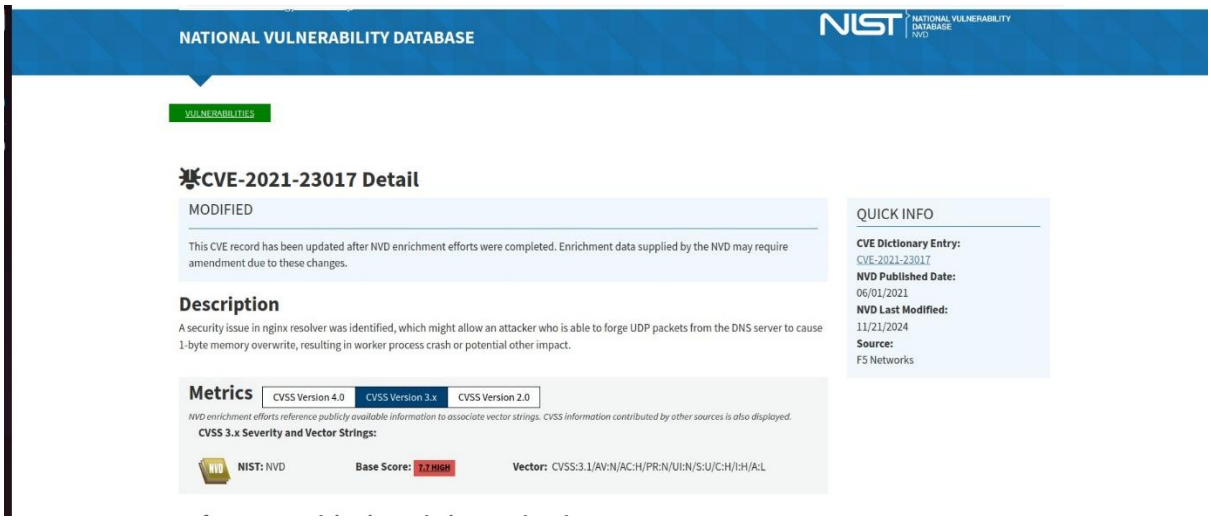
The **CVSS v3.0 Base Score of 7.5** indicates that this vulnerability is remotely exploitable, requires no authentication or user interaction, and has a high impact on availability. **The CVSSV3 impact score of 3.6** confirms that the primary effect is denial of service rather than data compromise, with no impact on confidentiality or integrity.

The **VPR score of 4.4** indicates a medium real-world threat level, as there are no known exploits and limited evidence of active exploitation, despite the technical severity.
EPSS value of 0.1395 suggests a modest likelihood of exploitation in the near term, meaning the vulnerability is not among the most frequently targeted issues but still represents a meaningful availability risk that should be addressed during routine patching.

D. Nginx 0.6.x < 1.20.1 1-Byte Memory Overwrite RCE



Further Details on NIST NVD Using CVE Tag



Nessus Description: According to its Server response header, the installed version of Nginx is 0.6.18 prior to 1.20.1. It is, therefore, affected by a remote code execution vulnerability. A security issue in Nginx resolver was identified, which might allow an unauthenticated remote attacker to cause 1-byte memory overwrite by using a specially crafted DNS response, resulting in worker process crash or, potentially, in arbitrary code execution.

NIST Description: A security issue in Nginx resolver was identified, which might allow an attacker who is able to forge UDP packets from the DNS server to cause 1-byte memory overwrite, resulting in worker process crash or potential other impact.

Vulnerability Publication Date: 25th May, 2021.
Patch Publication Date: 25th May, 2021.

Exploitation Method: CVE-2021-23017 can be exploited by an attacker who is able to forge UDP packets that appear to originate from the DNS server used by Nginx. By sending a specially crafted DNS response, the attacker can trigger a flaw in the Nginx resolver that results in a 1-byte memory overwrite within a worker process. This can cause the worker to crash or lead to memory corruption with potential confidentiality and integrity impact. Although the attack requires high complexity due to the need to spoof DNS responses, it is remotely exploitable without authentication or user interaction, and proof-of-concept exploit code is publicly available.

Exploit Available - True
Exploit Ease – Yes

Vulnerability ID as assigned by Nessus: 150154
Severity according to Nessus: High
CVE Code: CVE-2021-23017

For CVE-2021-23017, the **CVSS v3.1 Base Score of 7.7** reflects a remotely exploitable vulnerability with high impact on confidentiality and integrity if successfully exploited, despite the high attack complexity required to forge DNS responses.

The **CVSSV3 impact score of 5.5** confirms that the primary concern is data and code compromise rather than simple service disruption.

VPR score of 7.0 indicates that, in practical terms, this is a high-priority issue due to the availability of proof-of-concept exploit code and the widespread use of Nginx.

The **EPSS value of 0.7317** further suggests a high likelihood of exploitation, meaning this vulnerability is statistically very likely to be targeted and should be remediated as a matter of urgency.

Successful Ansible (patch automation software) Installation

```
ubuntu@ubuntu:~$ sudo apt install ansible -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ansible is already the newest version (9.2.0+dfsg-0ubuntu5).
0 upgraded, 0 newly installed, 0 to remove and 380 not upgraded.
ubuntu@ubuntu:~$ ansible --version
ansible [core 2.16.3]
  config file = None
  configured module search path = ['/home/ubuntu/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  ansible collection location = /home/ubuntu/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.12.3 (main, Aug 14 2025, 17:47:21) [GCC 13.3.0] (/usr/bin/python3)
  jinja version = 3.1.2
  libyaml = True
ubuntu@ubuntu:~$
```

Patch Script File Created with Nano

```
GNU nano 7.2                                    update_nginx.yml
---
- name: Patch Nginx on Ubuntu
  hosts: localhost
  become: yes

  tasks:
    - name: Remove outdated Nginx (if installed manually)
      file:
        path: /usr/local/nginx
        state: absent

    - name: Ensure system package lists are updated
      apt:
        update_cache: yes

    - name: Install latest Nginx
      apt:
        name: nginx
        state: latest

    - name: Start and enable Nginx service
      systemd:
        name: nginx
        state: started
        enabled: yes

^G Help      ^O Write Out   ^W Where Is    ^K Cut       ^T Execute   ^C Location   M-U Undo      M-A Set Mark   M-] To Bracket  M-Q Previous   ^B Back
^X Exit      ^R Read File   ^\ Replace     ^U Paste     ^J Justify   ^/ Go To Line M-E Redo      M-6 Copy       ^Q Where Was    M-W Next       ^F Forward
```

Successful Patch Execution and Updated Nginx Software (version 1.24.0)

```
ubuntu@ubuntu:~$ ansible-playbook update_nginx.yml --ask-become-pass
BECOME password:
[WARNING]: No inventory was parsed, only implicit localhost is available
[WARNING]: provided hosts list is empty, only localhost is available. Note that the
implicit localhost does not match 'all'

PLAY [Patch Nginx on Ubuntu] ***************************************************

TASK [Gathering Facts] *********************************************************
ok: [localhost]

TASK [Remove outdated Nginx (if installed manually)] ***************************
ok: [localhost]

TASK [Ensure system package lists are updated] *********************************
changed: [localhost]

TASK [Install latest Nginx] ****************************************************
changed: [localhost]

TASK [Start and enable Nginx service] ******************************************
ok: [localhost]

PLAY RECAP *********************************************************************
localhost                  : ok=5    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

ubuntu@ubuntu:~$ nginx -version
nginx version: nginx/1.24.0 (Ubuntu)
ubuntu@ubuntu:~$
```

7. Business Impact Analysis

- **Service Disruption:** DoS vulnerabilities could result in downtime
- **Data Exposure:** Information disclosure may leak sensitive data
- **System Compromise:** RCE vulnerability could lead to full server takeover
- **Reputational Damage:** Breaches may reduce client trust

8. Remediation Actions

8.1 Automated Patching with Ansible

Ansible was used to:

- Update repositories
- Install nginx 1.24.0
- Restart the service
- Validate installation

This ensured repeatable, auditable, and consistent patching.

8.2 SMTP Automation

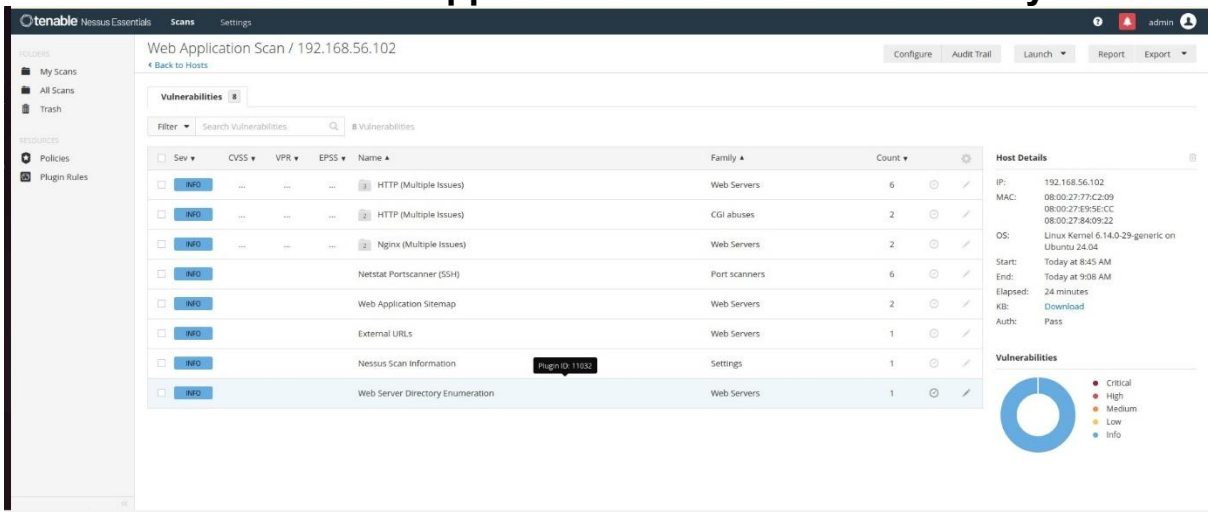
SMTP was configured to automatically email scan results to stakeholders, improving SOC visibility and overall security posture.

9. Post-Remediation Validation

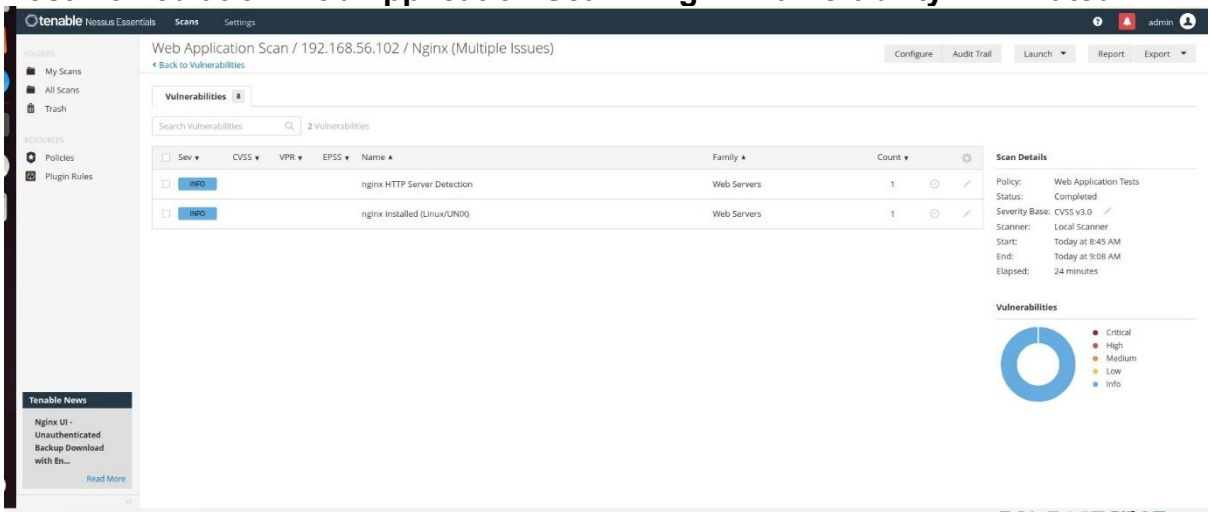
- Re-executed web application scan
- No vulnerabilities detected
- Only informational findings remained

This confirms successful remediation within the assessment scope.

Post-Remediation Web Application Scan – No Vulnerability Found



Post-Remediation Web Application Scan – Nginx Vulnerability Eliminated



10. Limitations

- Nessus may produce false positives or miss zero-day vulnerabilities
- Web scan was unauthenticated (limited coverage)
- Assessment focused only on Nginx and exposed services

11. Recommendations

- Implement continuous vulnerability scanning
- Enforce regular patch management cycles
- Enable authenticated web application scanning
- Deploy Web Application Firewall (WAF)
- Integrate vulnerability management with SIEM/SOC monitoring

12. Conclusion

The assessment successfully identified and remediated critical vulnerabilities affecting the Nginx server. Automation using Ansible significantly improved remediation efficiency and consistency.

The system now demonstrates a strong security posture within the assessed scope. However, continuous monitoring and proactive vulnerability management remain essential to maintaining long-term security resilience.